Using IPS data to understand the PTA Noise Budget

Work carried under the supervision of

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by

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Motivation

Pulsars: A rapidly rotating, highly magnetized neutron star that emits beams of electromagnetic radiation observed as periodic pulses when swepeed across Earth.

Pulsar Timing Arrays (PTAs): Networks of highly stable millisecond pulsars used as galactic-scale clocks to detect low-frequency gravitational waves via correlated timing residuals.

The gravitational-wave background (GWB): is observed in PTAs by detecting a common red-noise signal with a distinctive Hellings–Downs correlation pattern in the timing residuals of many millisecond pulsars.

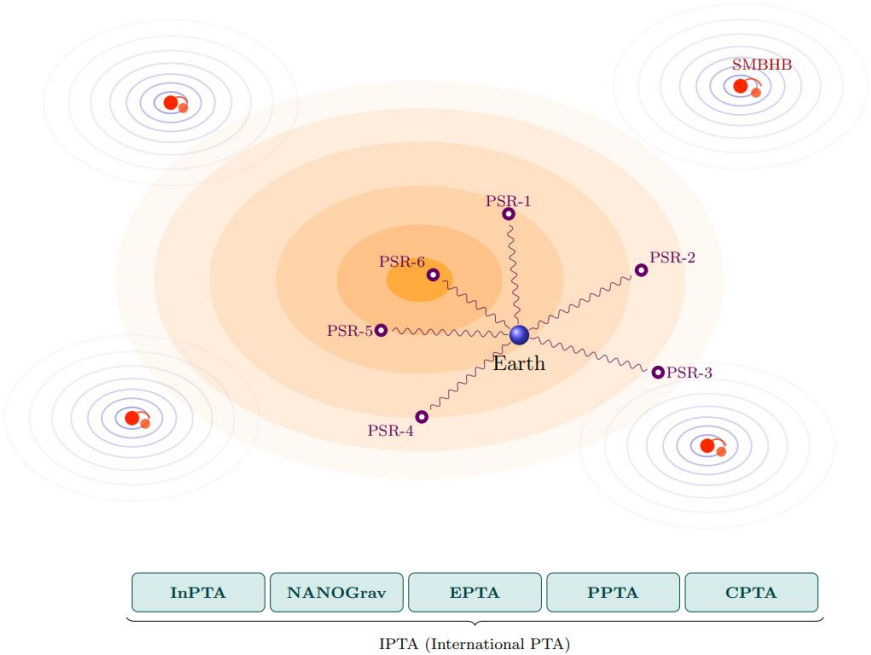


Fig: An Illustration of Pulsar Timing Arrays

Noise in PTAs

1. Radiometer Noise (White Noise)
2. Pulse Phase Jitter Noise
3. Spin Noise (Timing Noise)
4. Ephemeris Errors
5. Clock Errors
6. Dispersion Measure (DM) Variations
7. **Solar Wind Noise**
8. Instrumental / Backend Noise
9. **Gravitational Wave Background (GWB)**

Fig: Radiometer, GMRT (NCRA)

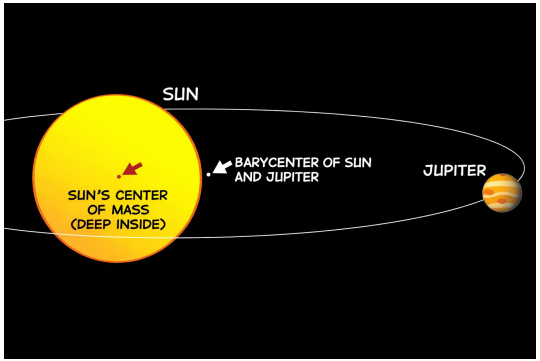


Fig: Ephemeris (Shifting SSB)

Fig: Solar Wind (SWOOPS)

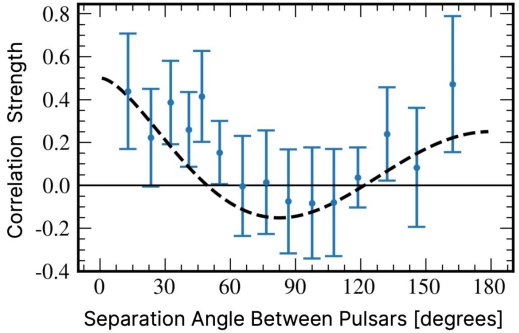


Fig: Gravitational-wave Background (NanoGrav)

The Solar Wind Challenge

Why theoretically modelling the sun is hard?

1. Strong Temporal Variability (Non-stationarity)
2. Spatial Anisotropy & Line-of-Sight Dependence
3. Multi-scale Turbulence (Kolmogorov-like but not exact)
4. Degeneracy with ISM DM Variations
5. Non-Gaussian
6. Transient Events

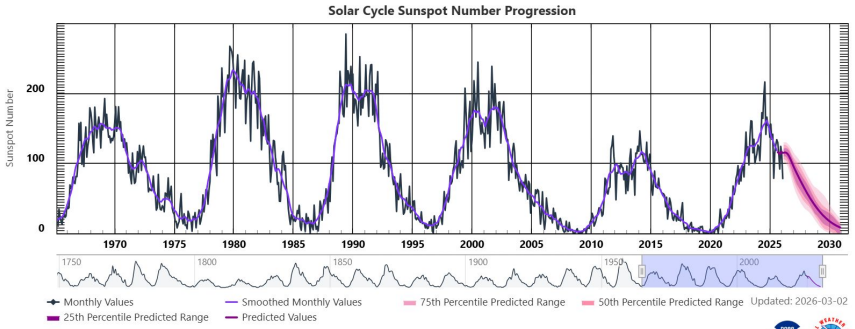


Fig: Solar Cycle from 1965-2026 (NOAA)

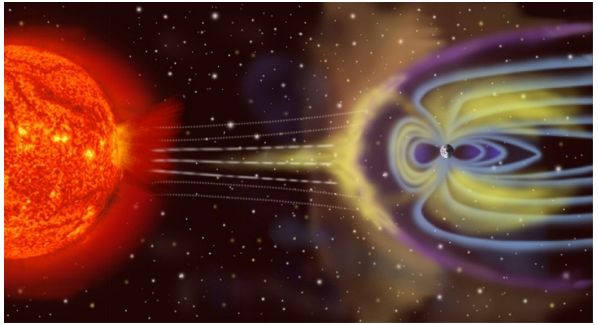


Fig: Solar Energetic Particles

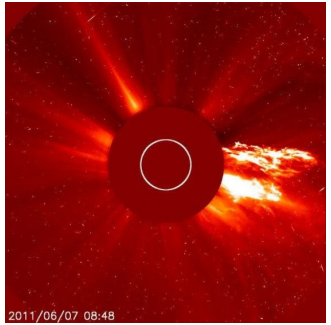


Fig: Coronal Mass Ejection (NASA)

What we do then? - Interplanetary Scintillation
Interplanetary scintillation (IPS) is the rapid fluctuation in the observed intensity of distant compact radio sources (**QUASARs**) caused by scattering from density irregularities in the turbulent solar wind.

Observation and Data Analysis

Telescope: Ooty Radio Telescope (ORT)

Located in southern India (Tamil Nadu's Nilgiri hills). The ORT is an equatorially mounted cylindrical paraboloid reflector, measuring **530m in length and 30m in width**. It operates at a **central frequency of 327 MHz** with a usable **bandwidth of 15 MHz**.

Fig: Ooty Radio Telescope (NCRA)

Data Analysis & Methodology

$$\textbf{Time Series} \rightarrow \textbf{FT} \rightarrow \textbf{PSD} \rightarrow \textbf{Scintillation Index}$$

1. Start with intensity time series: $I[n]$
2. Fourier Transform (FT)

$$\tilde{I}[k] = \sum_{n=0}^{N-1} I[n]\, e^{-2\pi i k n/N}$$

3. Power Spectral Density (PSD)

$$P[k] = \frac{\Delta t}{N} |\tilde{I}[k]|^2$$

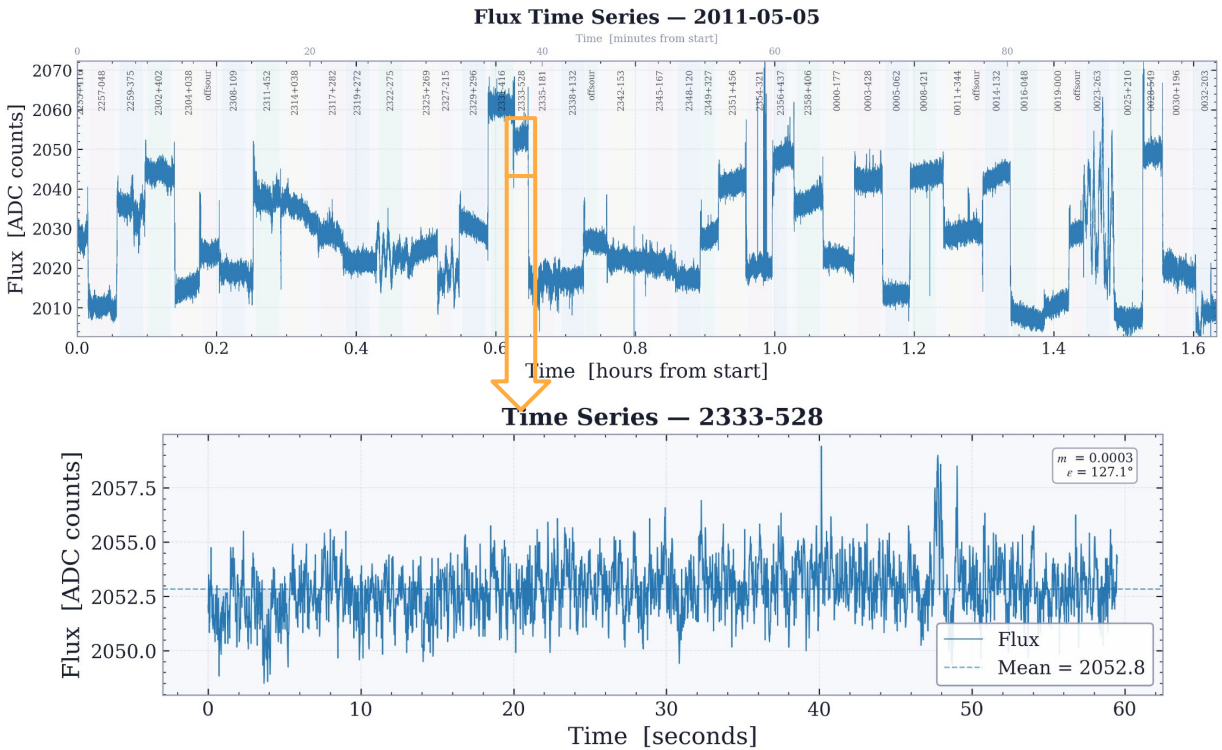
4. Noise floor subtraction: $P_{\mathrm{IPS}}[k] = P[k] - P_N$
5. Scintillation Index (integration over IPS band)

$$m^2 = \frac{1}{\langle I \rangle^2} \sum_{k=k_{\min}}^{k_{\max}} (P[k] - P_N)\, \Delta f\,, \qquad \Delta f = \frac{1}{N \Delta t}$$

Time Series Plots

A combined **intensity time series** for a **long observation block** consisting of several different IPS sources residing in a single observation campaign on 2nd May 2011.

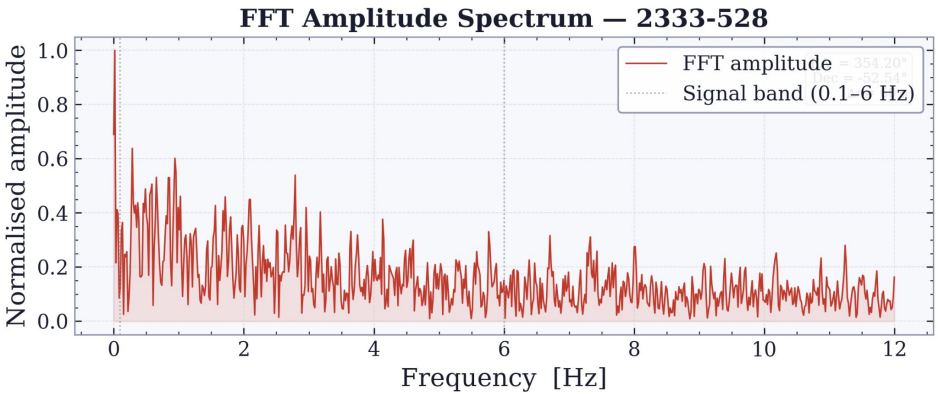
A cleaned **intensity time series** for a **single radio source**, showing the rapid fluctuations from interplanetary scintillation.



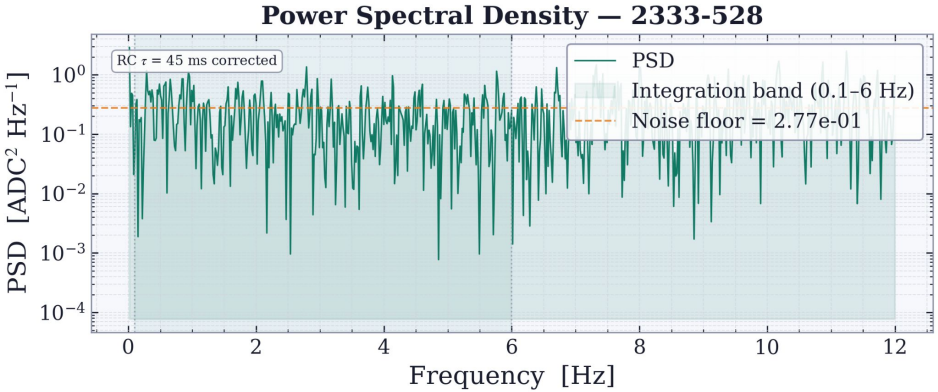
m = 0.0003 | ε = 127.1° | RA = 354.20° Dec = -52.54° | Δt = 60s

FT and PSD Plots

The **Fourier Transform (FT)** spectrum of the single source (2333-528) intensity time series.



The **Power Spectral Density (PSD)** on a log-log scaleof 2333-528. The section at the higher frequencies is the system white noise floor.



Sky Coverage & SI Distribution

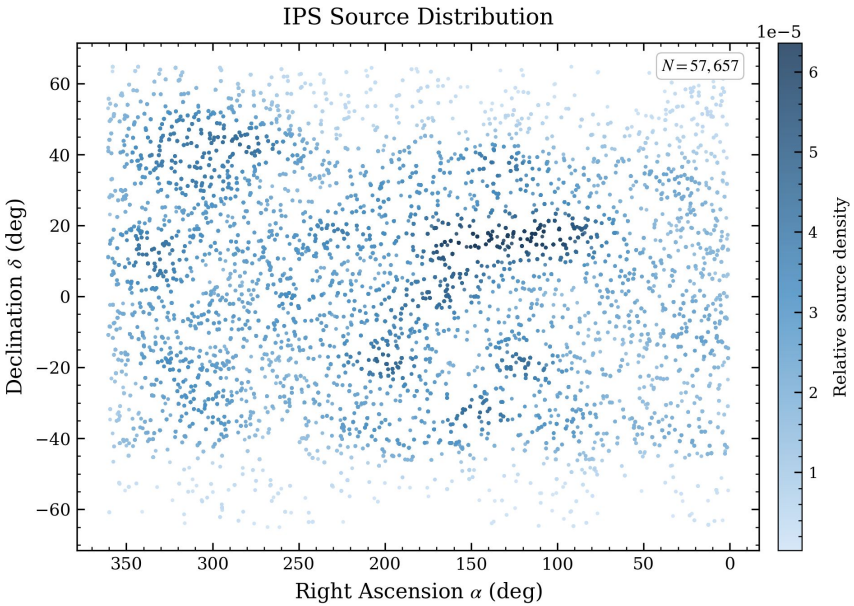


Fig: A plot showing the RA and Dec Coverage of the sources in the dataset

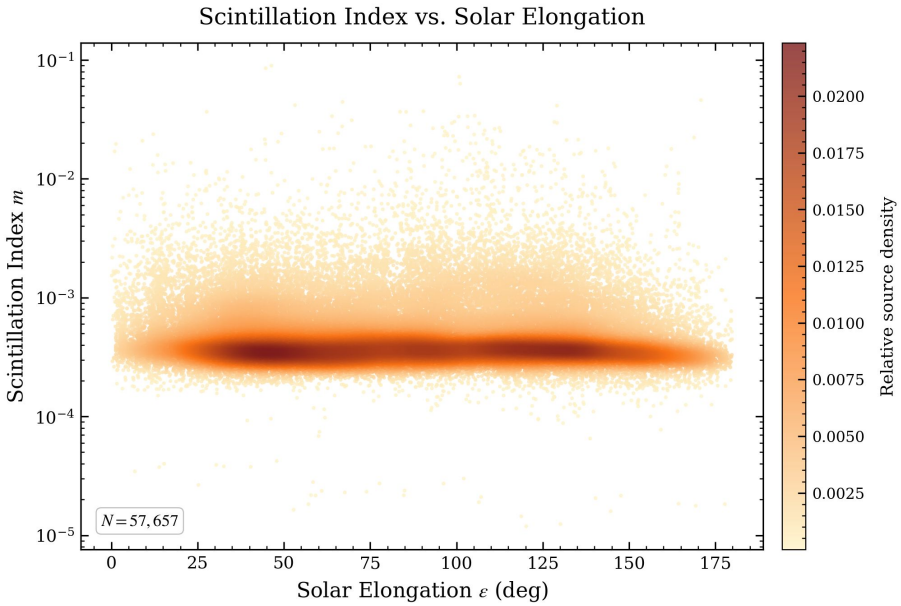


Fig: A plot showing SI value variation with Solar Elongation

Summarizing the IPS

We have successfully observed roughly 4,000 line of sights and 57,657 different IPS observation to capture the strength of electron density fluctuations. These observations are necessary to do Tomography, study transient events and calculate DM for PTA studies.

This video shows the celestial sphere with earth at the center and sun highlighted as a orange gradient patch on the sphere with dosts indicating QUASAR observations. The cmap shows the SI value measured there.

Vid: A video illustration of the observations being made by ORT in the month of May 2011.

Independent Detection of a Rare Event

On June 7, 2011, an M-2 flare occurred on the Sun which released a very large coronal mass ejection (CME). Much of the ejected material is much cooler (less than about 80,000K) and therefore appears dark against the brighter solar disk.

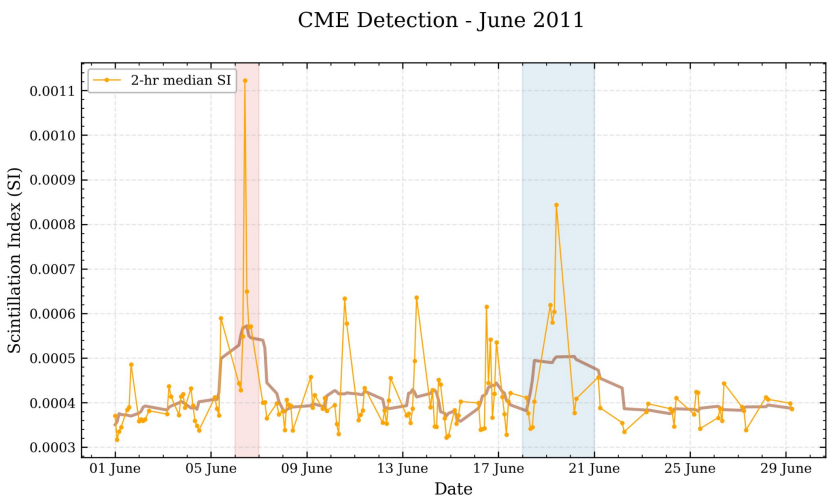
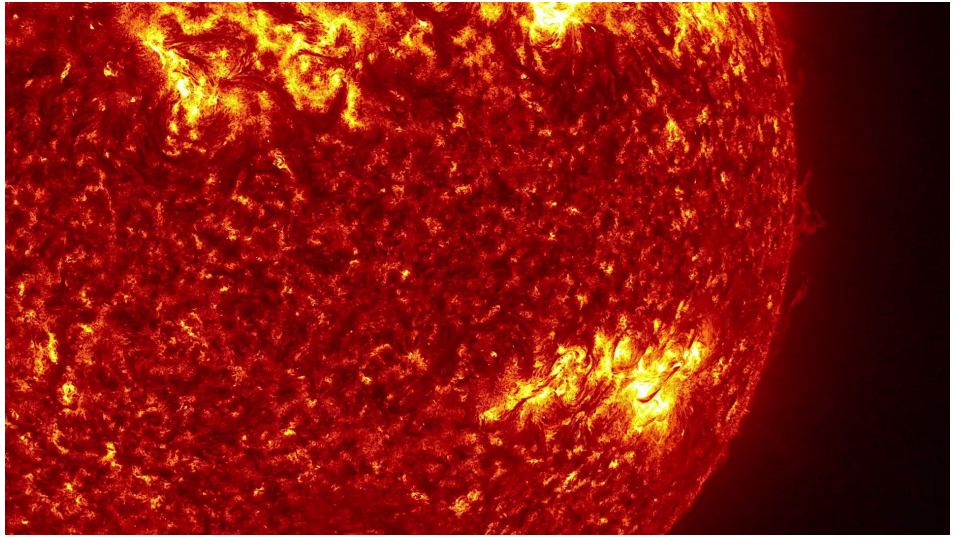


Fig: An independent detection of a rare event on June 7, 2011 captured by our study.



Vid: 7 June 2011: An M2 flare launched a massive CME with unusually cool (<80,000 K) plasma, appearing dark in SDO video (credit: NASA).

Tomography of the Heliosphere

3D tomographic reconstruction. By isolating the SIRT solver to monthly batches, we observe the large-scale shifting of density structures and the breakdown of the bimodal solar wind during the rising midway-phase of the solar cycle.

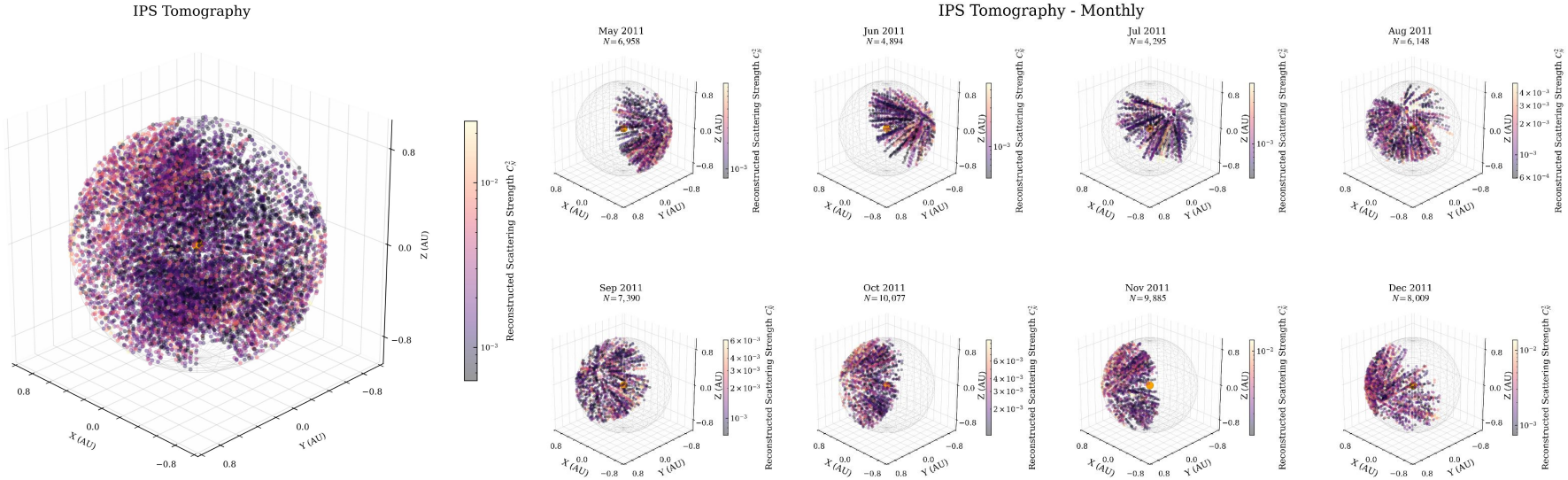


Fig. 3D reconstruction of IPS data from the ORT. The **left** panel shows the full eight-month volume, while the **right** panels show individual monthly reconstructions.

Conclusion and Future Work

The Problem: Standard solar wind models fail to capture heliospheric state that create noise in PTAs affecting gravitational wave background measurements.
The Approach: Utilized IPS data from ORT during the rising phase of Solar Cycle 24 in 2011.

Key Achievements:

- **Developed a robust data reduction pipeline**, converting telescope voltages into reliable SI measurements .
- Successfully detected sharp, anomalous spikes caused by events like the **June 7, 2011 CME**.
- **Developed the tomographic framework**, fed the data, and produced 3D reconstructions of the inner heliosphere.

Future Work:

- **Kinematic Tomography:** Upgrade the current static model to track the real-time kinematic propagation of CMEs.
- **Long-Term Solar Mapping:** Apply the pipeline to 33-year archive to build a highly accurate solar wind models.
- **Dynamic DM Corrections:** Integrate these density maps to get Dispersion Measure corrections for PTA's GWB detection.

Thank you for Listening !

Acknowledgements:

I would like to that Dr. Divya Oberoi for validating the results and providing the methodology.

I would also like to thank Dr. MA Krishnakumar for providing with the data.

Millisecond Pulsar, GWB and HD Correlations

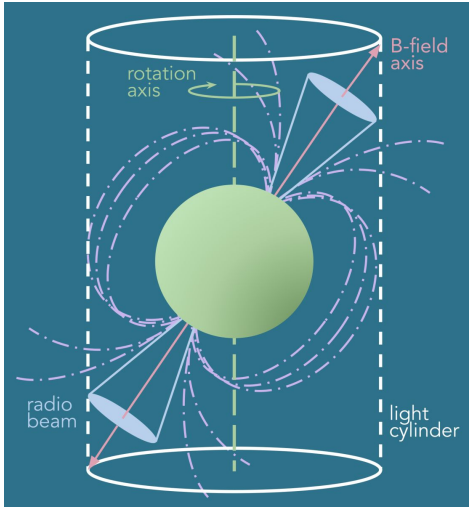


Fig: An illustration of a pulsar (credit: NANOGrav)

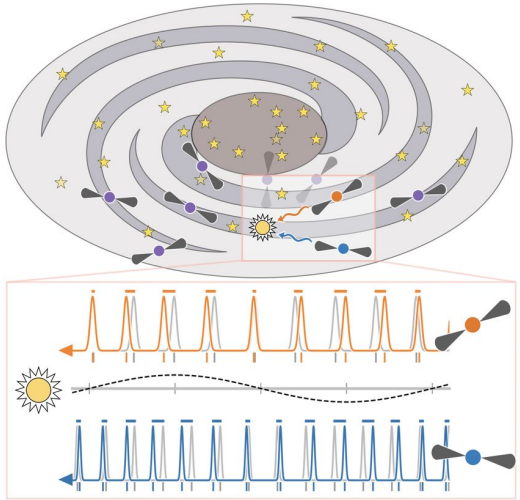


Fig: Illustration of PTAs (Source: Luke Zoltan Kelley, 2025, Pulsar Timing Arrays)

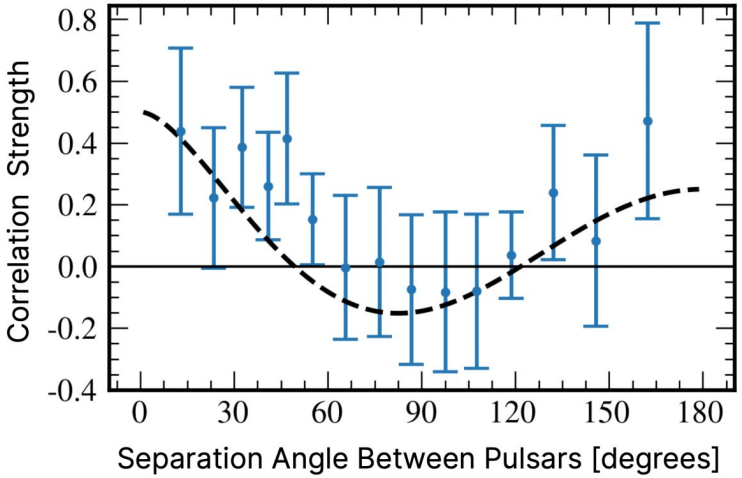


Fig: HD correlation plot (credit: NANOGrav)

A Rare Coronal Mass Ejection – 7th June 2011

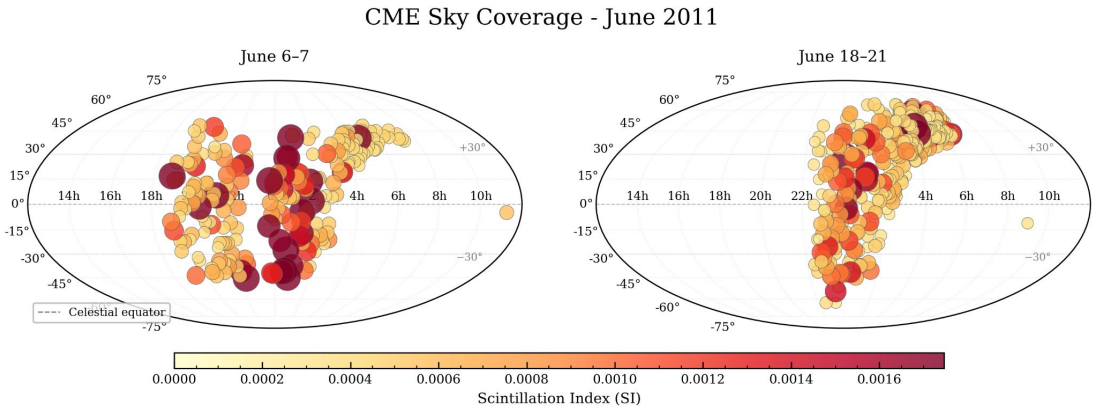
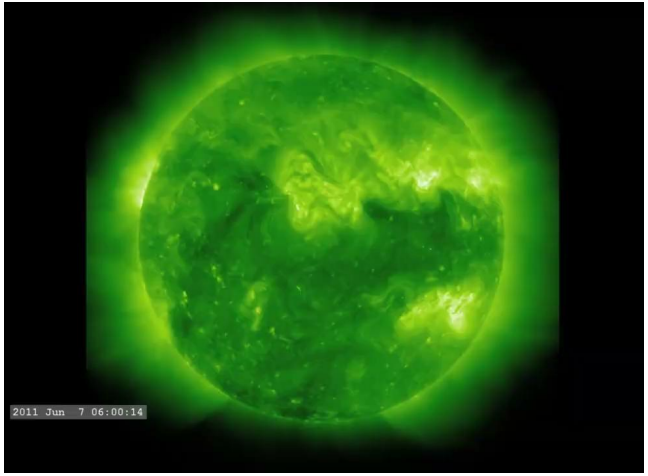


Fig: Mollweide projection sky maps corresponding to the periods for CME detection.



Vid: A huge CME on June7, 2011, extreme ultraviolet (EUV, 171A) (credit: NASA)